



Mini-project

Idea

- Prepare a complete technical implementation of an analytical data platform that follows a data warehouse architecture.
 - In particular, this platform should be able to tackle all of the identified OLAP needs of its users (through a set of reports, visualisations and dashboards) using the available data (originating from external sources/systems; available in different formats).
- An underlying assumption is that an intermediate storage solution, located between data sources and users, is utilised.
 - This storage is loaded with data, originating from external source systems, through a dedicated ETL solution.
 - ETL process involves staging extracted data in a separate relational database system.
 - Further the analytical storage is designed using the multidimensional model and is further limited to a single data mart solution, covering only a selected single line of business.
 - Implementation of the data mart requires a creation of a single dimensional data cube (using SSAS).
- Finally, user requirements are fulfilled through a set of well-designed and implemented reports, visualisations and interactive dashboards.



GENERAL OVERVIEW

GENERAL OVERVIEW OF 1ST MINI PROJECT PART:

- Focus is on Analysis
- Find dataset and get yourself familiar with the data, perform basic data profiling
- Describe analytical needs against selected dataset, determine dimensions, fact, and measures

GENERAL OVERVIEW OF 2ND MINI PROJECT PART:

- Focus is on Design
- Design multidimensional model for the selected dataset
- Create Logical Data Map
- Plan and design the ETL process: extract, clean, transform, and load your data

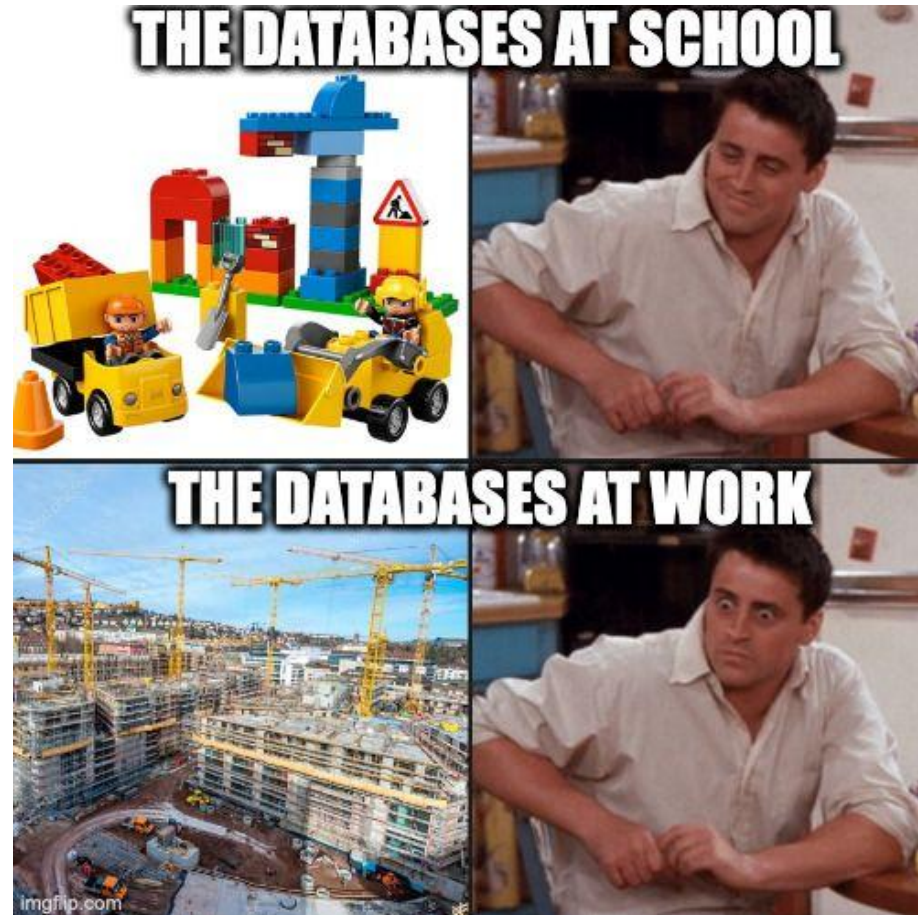
GENERAL OVERVIEW OF 3RD MINI PROJECT PART:

- Focus is on Implementation
- Implement the ETL process using SQL Server Integration Services SSIS
- Implement a multidimensional cube using SQL Server Analysis Services SSAS

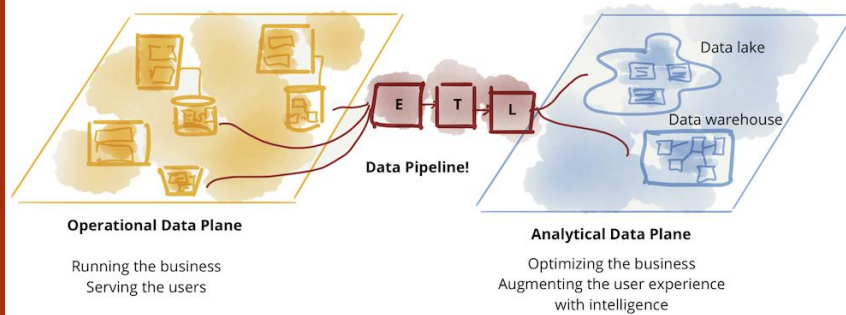
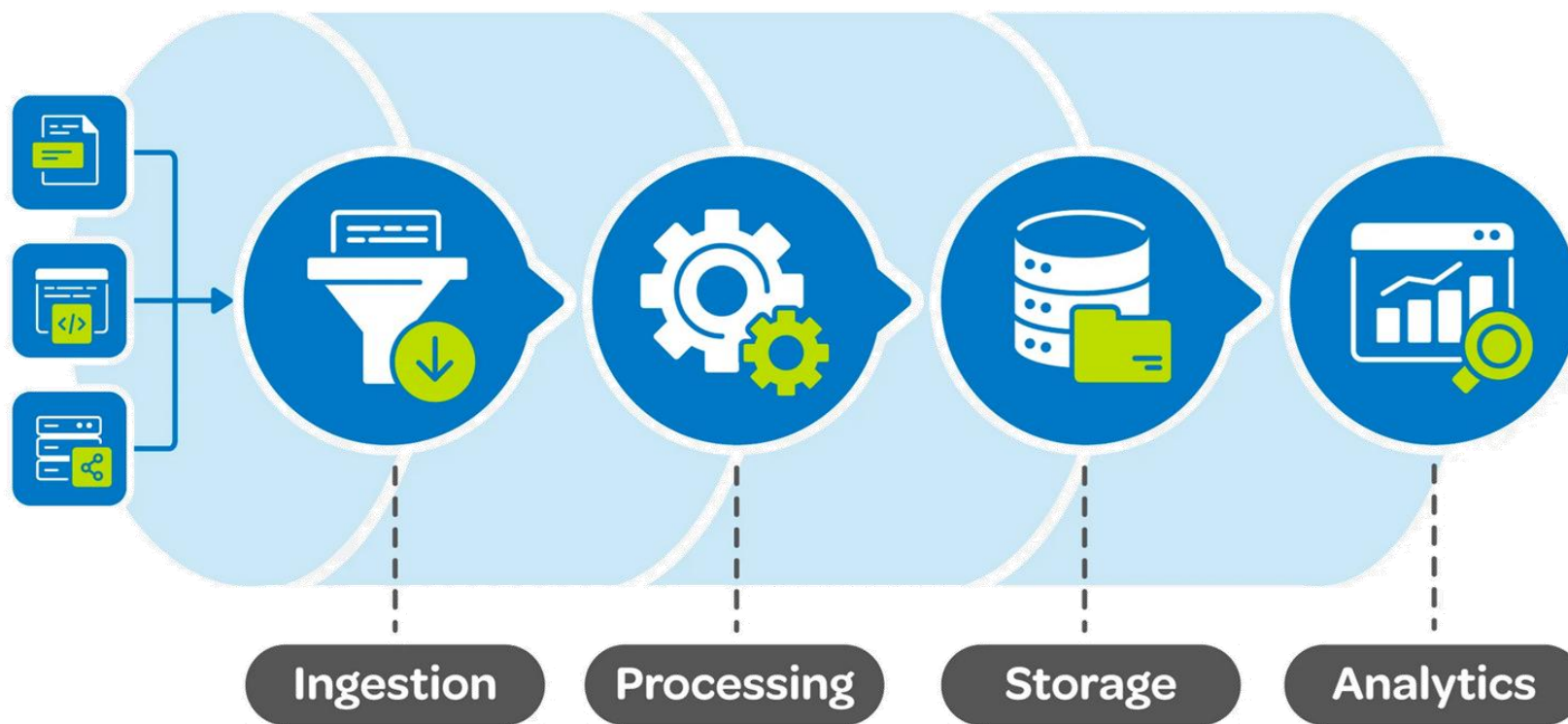
GENERAL OVERVIEW OF 4TH MINI PROJECT PART:

- Focus is on Usage
- Using the implemented cube perform basic data analysis and visualisation using Tableau/Power BI Desktop

Mini-project



- Do not overcomplicate your project
 - It is still a mini-project
 - You are encouraged to experiment – both tool-wise and data-wise
- We are working in a safe environment
 - Errors, mistakes, downtimes cost us nothing
 - Focus is on better understanding of different processes, architectures and technical solutions
- End-to-end
 - We want to tackle the entire process – from data sources to data consumption



 **Datamation**

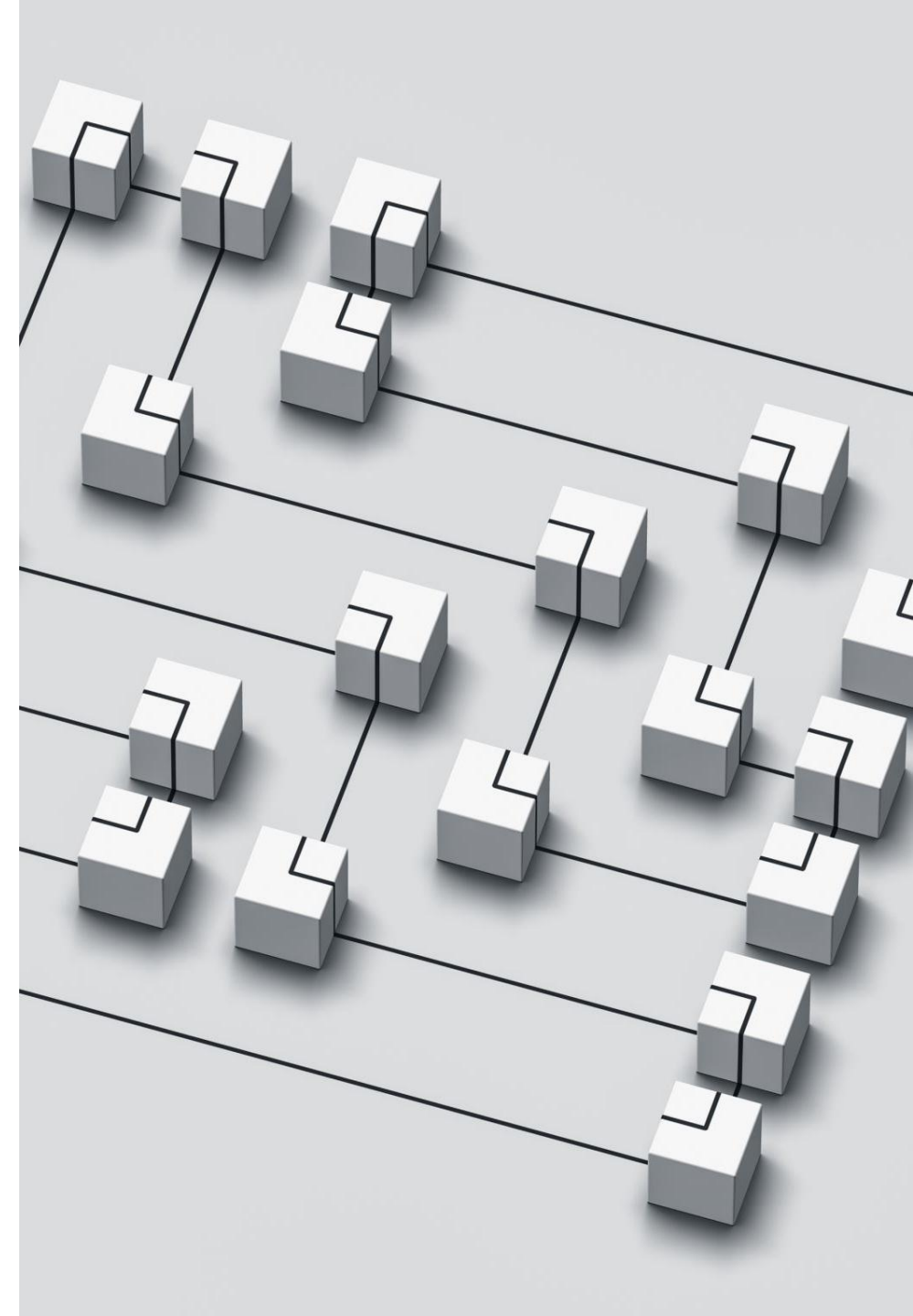
Plan

- Lab project:
 - 4 stages:
 - **Analysis**
 - **Data selection, requirement analysis**
 - Design
 - multidimensional model design
 - ETL design
 - basic cube – dimensional design
 - Implementation
 - ETL processing - implementation using SSIS
 - Dimensional implementation using SSAS
 - OLAP
 - Basics of data reporting, visualisation and analysis using Tableau
 - Using own datasets



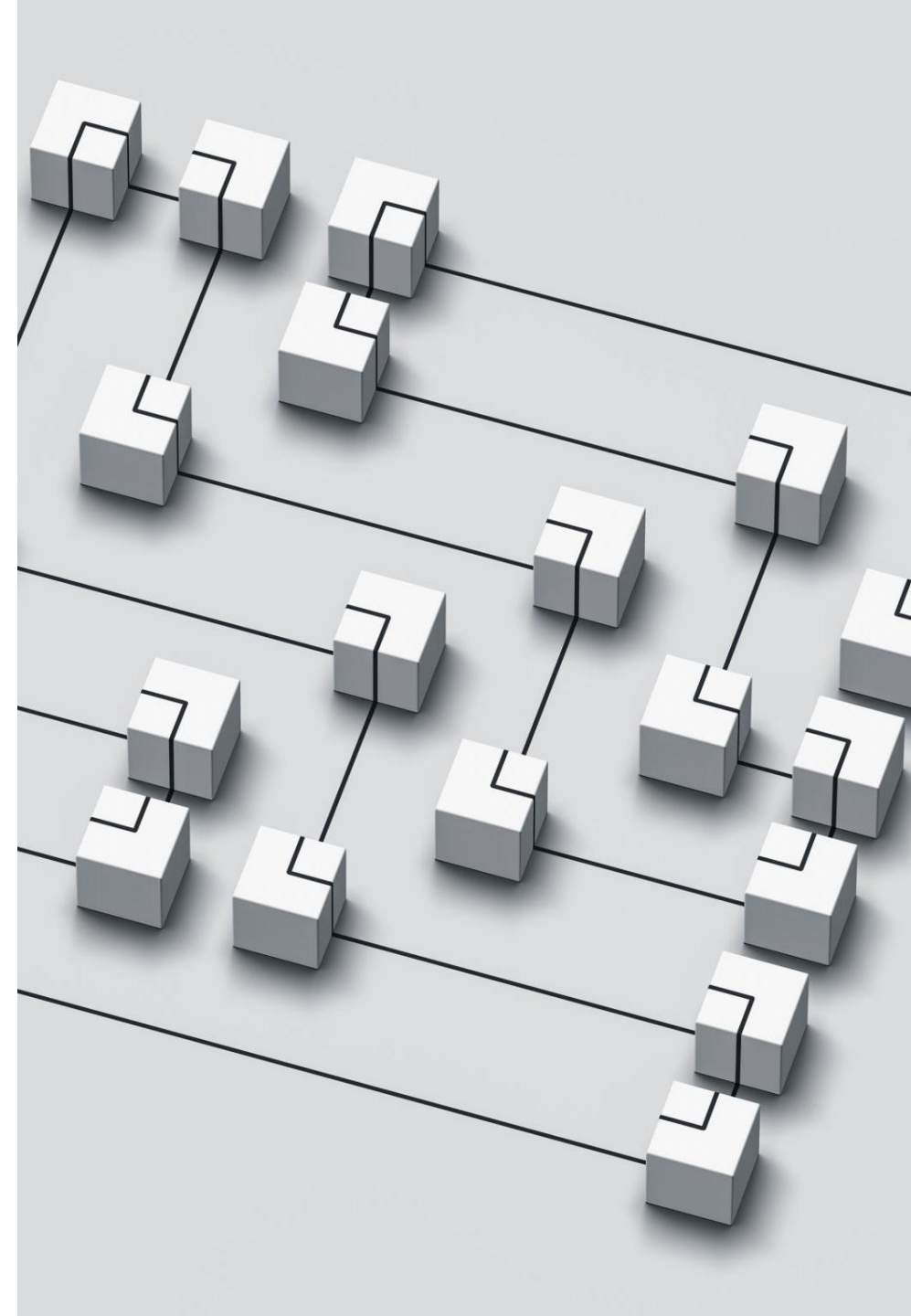
Data

- Remember to find/select a database/data source(s) for your mini-project.
 - The data should allow for multidimensional analysis (OLAP)
 - in the end you'll be building a data cube
 - As such, the data should be operational in nature and as close to data generation as possible
 - Try keeping the data close to its source
 - It would be good if you could see the data as a representation of a series of measurements (preferably quite a long one)
 - From which you can further take snapshots (preferably detailed) regarding the states of a particular business process,
 - events (understood in a quite generic way), that can be measured (using certain numerical values, eg. sales) and analysed from multiple perspectives (dimensions, eg. client, product, etc.). It
 - what triggers a snapshot might be of a different nature – event occurrence, time period, etc.



Data

- Examples
 - of events include: product sales, room rental, social net. comments, news posting, product returns, meeting attendance, item review, etc.
 - of measures include: sales amount, rental rate, comment's sentiment score, ad. revenue, processing time, late time, review score, etc.
 - of perspectives include: Product, Room, User, News outlet, Employee, Meeting, Item, etc.
- Bear in mind that the selection of your dataset does affect your project's overall quality and as such can affect your final score.
 - Although this should be a mini-project (data mart) and not a corporate class data warehouse design.





Data

- Data source characteristics to look for:
 - Look close
 - Domain you are interested in
 - Domain you can easily understand and play with
 - Originating from – at least – two data sources
 - e.g. NYC taxi rides and NYC bike share, sales data and weather data.
 - Operational data
 - Stay away from highlevel aggregates or highly processed analysis-ready data
 - Data should have enough attributes
 - Available subjects should be rich-enough
 - Data can be a series of flat files (csv, tsv, txt), data dumps, database (any database), or other
 - a mixture of both is encouraged



Data

- Data
 - Doesn't have to be huge
 - should be „rich enough” to support the project's goals
 - 4-5 dimensions with at least 6 attributes each
- Possible sources
 - Existing datasets – free and private
 - API consumption / Web scrapping
 - As easy as retrieve the data, as complex as using ML models
 - Simulation
- Set a nice application theme

- Each of the selected dataset should originate from – at least – two data sources:
 - e.g., NYC taxi rides and NYC bike share, shipment and local weather, POS sales and geopolitical context.
 - It is possible to focus on data from a single source, but this needs to be properly justified and this might negatively affect your score.
- Data can come in different formats from different systems.
 - For instance, it can be a series of flat files (csv, tsv, txt) – like an extract from a web service, dump from a database, etc. – or online/offline access to databases servers (any DBMS is acceptable) – though a mixture of different formats is highly encouraged.
- Please note, that data should have enough attributes (should be wide enough) – both continuous and categorical are possible.
 - The idea is that your model should be able to cover of at least 4 measures (3 is a minimum for a basic score), at least 4 dimensions (4 is minimum for a basic score) with at least 5-10 attributes each.
- You cannot use AdventureWorks database for this task!



- Data example
 - <http://iot.ee.surrey.ac.uk:8080/datasets.html>



Description	Duration	Location (Provider)	Type
Road Traffic Data	2/2014 - 6/2014 8/2014 - 9/2014 10/2014 - 11/2014 07/2015 - 10/2015	Aarhus, Denmark (Open Data Aarhus)	Real
Pollution Data	8/2014 - 10/2014 2/2014 - 6/2014 8/2014 - 9/2014	Aarhus, Denmark (Open Data Aarhus) Brasov, Romania Brasov, Romania	Generated
Weather Data	2/2014 - 6/2014 8/2014 - 9/2014 2/2014 - 6/2014 8/2014 - 9/2014	Aarhus, Denmark (Open Data Aarhus) Aarhus, Denmark (Open Data Aarhus) Brasov, Romania Brasov, Romania	Real
Cultural Event Data	5/2014 - 1/2015	Aarhus, Denmark (Open Data Aarhus)	Real
Social Event Data	6/2012 - 6/2014	Surrey, UK (Municipality RSS)	Real
Library Event Data	10/2013 - 6/2015	Aarhus, Denmark	Real
Parking Data	5/2014 - 11/2014	Aarhus, Denmark	Real

- Data example
 - <https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>



Data Dictionaries and MetaData

- Trip Record User Guide
- Yellow Trips Data Dictionary
- Green Trips Data Dictionary
- FHV Trips Data Dictionary
- High Volume FHV Trips Data Dictionary

Taxi Zone Maps and Lookup Tables

- Taxi Zone Lookup Table (CSV)
- Taxi Zone Shapefile (CSV)
- Taxi Zone Map – Bronx (JPG)
- Taxi Zone Map – Brooklyn (JPG)
- Taxi Zone Map – Manhattan (JPG)
- Taxi Zone Map – Queens (JPG)
- Taxi Zone Map – Staten Island (JPG)

- Many, many, many options
 - Kaggle
 - Data.world and data portals
 - Knoema
 - r/datasets
 - data.gov
 - dane.gov.pl

Presentation

- The idea of the presentation is to present your data sources and sketch possible user(s) to the rest of the group.
- Your presentation should focus on (not limited to):
 - domain introduction (overview domain, introduce basic concepts and notions),
 - user type specification and overview (overview different types of end users – better focus on 1-2),
 - data source details (access and technical details),
 - dataset overview (content overview, model overview, granularity) and availability (format, type, change rate),
 - dimensional synopsis (identify and name underlying business process (events/statuses) and possible dimensions (context))
 - selected user needs (identify selected specific user needs, focus on decision support)
 - comment on expected results (highlight, end-user oriented).

Presentation

- REMINDER:
 - Your dataset should originate from two separate sources
 - e.g., sales orders and accompanying weather data, twitter feeds and stock prices, crime events and alcohol consumption.
 - Note – if you have a more complex and broader idea of a single data source, please consult it before making final decisions
 - Your group (of 2) should initially have two data sources
 - Then select one, as the preferred one, and present to the group; the other dataset should be at least highlighted in your presentation
 - The only limit is 10-15min. for the presentation depending on the number of groups in the class.
 - There is no enforced format of your data source
 - However, you are encouraged to use heterogenous formats